

Dao Xuan Tan

Data Engineer

📍 Ho Chi Minh, Vietnam

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Summary

Data engineer with 1+ year building and operating production **Python ETL and batch pipelines** over **financial market and time-series data**, from ingestion and transformation through **PostgreSQL** persistence, **Kafka** streaming, and scheduled containerized workflows with stage dependencies, retries, and failure alerting. Strongest on **SQL, data-quality validation**, and production monitoring.

Experience

Pi Associates, Software Engineer (Data & Quant Platform)

Ho Chi Minh

Python, SQL, PostgreSQL, Kafka, Redis, pandas, Docker, Kubernetes, GCP, Grafana

Jan 2026 – present

- Own core development of an internal **Python** data and backtesting platform serving **20** quant researchers, running **tens of thousands of jobs daily**
- Built the batch **ETL workflow** end to end, ingestion of **financial market and time-series data**, transformation, metrics computation, **reconciliation across runs**, and **report generation**, persisting to **PostgreSQL** with schema design and raw **SQL**, plus **Redis** caching
- Orchestrated recurring and live workloads as scheduled **Kubernetes** jobs across **GCP** and on-premises with **stage dependencies, retries, and failure alerting**, plus **Grafana** monitoring and input validation that cut failed runs and manual debugging
- Detected nondeterministic output across identical runs, traced it to a failed deployment serving incorrect data, and added the **data-quality validation and alerting** that now catches that class of failure before results reach users
- Built and operated a **Kafka** producer streaming trading signals in real time to downstream consumers
- Cut common **1–5 GB** workloads from minutes to **under 10 seconds** through parallelism and vectorization, and resolved a memory leak that cut peak memory by **an order of magnitude**

AcamarAI, AI Engineer

Ho Chi Minh

Python, SQL, ingestion pipelines, classification, entity resolution, evaluation

June 2025 – Jan 2026

- Built ingestion pipelines over **millions of records**, parsing, validating, and normalizing structured and unstructured documents into a self-hosted store
- Built **classification** and **entity-resolution** workflows combining deterministic rules with **LLMs**, with automated evaluation and human review to catch quality regressions
- Owned **3** core product features from requirements through rollout for **Fortune 500** beta users

Skills

Data Engineering: Python (pandas, asyncio, multiprocessing), SQL, ETL and batch pipelines, ingestion and normalization

Databases & Modeling: PostgreSQL (schema design, raw SQL), Redis, relational and time-series data modeling

Workflow & Streaming: Kubernetes job orchestration (DAG-style stage dependencies, retries, trigger rules, failure branches), Airflow (working knowledge: KubernetesPodOperator, sensors, XCom, catchup/backfill), Kafka streaming

Data Quality & Testing: Input and schema validation, correctness and reconciliation checks, anomaly and silent-failure detection, pytest (unit and integration), Grafana monitoring and alerting

Infrastructure: Docker, Kubernetes, GCP, on-premises, Linux, GitHub Actions CI/CD, Git

AI & LLM: LLM-assisted data processing, RAG and embedding pipelines, self-hosted model serving (vLLM), output evaluation

Languages: Vietnamese (Native), English (IELTS 6.5)

Education

BSc **Ho Chi Minh University of Science**, Mathematics and Computer Science (Data Science)

Ho Chi Minh City

- GPA: 8.74/10.0 (3.74/4.0). Expected graduation **Dec 2026**, working full-time since June 2025

Oct 2022 – Dec 2026